Aden James Calinisan

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Embedded Systems Engineering | C/C++ • Zephyr RTOS • SPI/I²C • MCU/NPU Integration

EDUCATION

University of California, Irvine (GPA 3.846) **December 2026**
Bachelor of Science, Computer Engineering
Florida State College at Jacksonville (GPA 4.000) **May 2023**

SKILLS

- **Programming:** C, C++, Python (Numpy, Pytorch, Tensorflow), Zephyr RTOS, Java, Linux, GitHub, GitLab
- **Software:** Jupyter Notebook, VSCode Extensions, Arduino IDE, Microsoft Excel, Solidworks, MATLAB, OrcaSlicer

WORK EXPERIENCE

Applications Engineering Intern | *Syntiant - Irvine, CA* **February 2026 – Present**
- Integrate NPU hardware with Zephyr RTOS using SPI and I²C communication protocol for enhanced portability
- Benchmark embedded AI/ML workloads across multiple MCU and NPU platforms (ie. Alif, NXP, Nordic, Silabs)
- Create an easy-to-read Excel spreadsheet that clearly compares AI benchmarks (uJ/inference, inferences/sec, etc)
- Operate multiple VSCode extension workflows for building, flashing, and debugging embedded systems projects
- Navigate SRAM constraints, NN model compiling, and MLPerf Tiny implementation on various microcontrollers
- Debug with Oscilloscope to validate I²C communication and confirm clock frequencies

Barista & Finance | *Boba Bee - Lake Forest, CA* **January 2024 – February 2026**
- Learn to communicate with event coordinators and delegate tasks while heading up catering events
- Create quarterly financial reports for 4 Boba Bee locations cutting accounting fees by 80%

Finance for Turo Business | *KayCal Ventures LLC - Jacksonville, FL* **July 2024 – November 2024**
- Assist the #1 Turo cohost (by fleet size and earnings) with payroll at *RentNiceCars.com*.
- Reduce payroll processing time by 75% for 23 virtual assistants using Microsoft Excel.

Electrical Engineering Intern | *The Haskell Company - Jacksonville, FL* **June 2022 – July 2022**
- Communicate professionally among engineers and learn basics of electrical design and project documentation

PROJECTS

Soccer Rover **June 2025 – September 2025**
Developed for High School Robotics Camp at UC Irvine Samueli School of Engineering

I was entrusted with the full responsibility of creating and teaching an autonomous soccer rover project. I built the project with Raspberry Pi 3, 3D prints, a laser cut chassis, motors, a power system, and a camera module. Open CV was used for ball detection and Python was used for the rover's directional logic. Within one week of development, 28 students had their very own soccer rover to bring home. (See Github for more)

Optical Neural Network Research **May 2025 – February 2026**
Shcherbakov Nanophotonics Lab at UC Irvine

Researching the translation of Convolutional Neural Networks from slower digital computation to a speed of light optical neural network using Spatial Light Modulation in a 4f System along with nonlinear optical responses.

Fitness Tracker **January 2024 – April 2024**
Engineering 7B Class at UC Irvine

My team and I developed a wearable Fitness Tracker. The tracker communicates via WiFi and the BlynkIoT phone app. I led soldering and assembling the bus lines, accelerometer, barometer, battery, and WEMOS development board.

Open Project Space **September 2023 – June 2024**
IEEE extracurricular class at UC Irvine

In this extracurricular course, I completed eight embedded systems projects. As a capstone, I built a remote-controlled rover with NRF24L01 radio, Arduino Nano, L293D Driver, and a custom PCB design.

Personal Activities: Piano, Volleyball, Tennis, Biking, Volunteering at Church, Raspberry Pi Tinkering, Fighting Games

github.com/ajcalinisan · linkedin.com/in/aden-calinisan